Project 2 rubric

Mile High Stays

This rubric evaluates the analysis document and one-page executive summary. It rewards correct, reproducible analysis, clear explanations, and business decisions supported by evidence. The total is 100 points.

Criterion	Maximum points
Analysis	20
Visualization quality and accuracy	10
Interpretation and limitations	15
Business decision and recommendation	10
Data preparation and reproducibility	15
Technical report communication	15
Executive summary	15
Total	100

Scoring

Assign each criterion an integer rating: **4** exceeds expectations, **3** meets expectations, **2** partially meets expectations, **1** falls below expectations, or **0** when there is no assessable evidence. Criterion points equal its maximum points multiplied by the rating divided by 4. Add the criterion points and round only the final total to one decimal place.

For example, a rating of 3 on Analysis earns 15 of 20 points. Ratings of 3 on every criterion yield 75 of 100 points. Use 0 for missing work rather than automatically awarding the lowest nonzero rating.

Using the criteria

The analysis should address the Denver pricing model, client-property recommendations, and comparisons with Seattle and Portland. Appropriate methods from lectures 01-11 can earn full credit; advanced techniques beyond the course are not required.

The rendered Word or PDF analysis document must contain the code needed to reproduce its findings. A separate QMD file is useful but is not a submission requirement; do not penalize its absence. Ignore harmless startup messages when execution and results are sound.

Score each defect primarily under the criterion it affects. Distinguish incorrect calculations from inaccurate interpretation, and both from unclear presentation. Avoid repeated deductions for the same underlying mistake; document separate problems when they warrant separate judgments.

Analysis and visualization

Analysis - 20 points

Rating	Evidence in the submission
4	Addresses all three client requests with clearly defined outcomes and comparisons. Models, predictions, calculations, and methods are correct and sufficient for the decision; sample sizes and relevant characteristics inform the analysis. Completes every step of the inferential workflow shared in class.
3	Addresses the assigned objective using appropriate methods and mostly correct comparisons. Definitions or supporting evidence have minor gaps that do not materially change the main findings. Completes all essential steps of the inferential workflow shared in class but omits at least one minor step.
2	Addresses only part of the objective, leaves important groups or metrics unclear, or makes errors that weaken a key comparison or finding. Completes some steps of the inferential workflow shared in class but omits or inadequately completes at least one essential step.
1	Uses an unassigned objective, inappropriate methods, or major calculation errors; the work provides little usable evidence for the assigned decision. Omits most essential steps of the inferential workflow shared in class or does not follow a coherent inferential workflow.

Visualization quality and accuracy - 10 points

Rating	Evidence in the submission
4	Uses appropriate, fully rendered figures with readable titles, axes, units, legends, and relevant group sizes. Rates and denominators are correct; scales and comparisons communicate the findings without misleading the reader.
3	Provides readable, appropriate plots that support the analysis. Minor labeling or formatting issues do not change their substantive interpretation.
2	Includes missing labels, unclear units or denominators, hard-to-read elements, or unsuitable chart choices that obscure a meaningful comparison.
1	Omits the required plots or uses substantially broken or misleading visuals that prevent accurate interpretation of the main evidence.

Interpretation and recommendations

Interpretation and limitations - 15 points

Rating	Evidence in the submission
4	Explains what the results mean accurately, identifies limitations that affect the decision, and distinguishes observed patterns from causal effects or forecasts. Conclusions remain consistent with the evidence and group sizes.
3	Interprets the main findings correctly and notes the most important limitation. Some implications or qualifications could be clearer without materially overstating the result.
2	Gives superficial explanations, omits a material limitation, or makes an unsupported claim that affects how a reader would use the result.
1	Misinterprets central results or presents unsupported causal, time, or predictive claims as established findings.

Business decision and recommendation - 10 points

Rating	Evidence in the submission
4	Proposes a concrete action for identified properties or pricing decisions, explains how the evidence supports it, and considers scope and practical tradeoffs. Provides an appropriate test or next step with a measurable outcome.
3	Proposes a relevant, feasible action tied to the main findings. The audience or next step is identifiable, but the rationale or success measure could be more specific.
2	Offers a generic action, weakly connects it to the findings, or leaves the target audience and implementation or validation step unclear.
1	Provides little usable recommendation, or recommends action that conflicts with the evidence or assigned business objective.

Preparation and technical reporting

Data preparation and reproducibility - 15 points

Rating	Evidence in the submission
4	Starts from the supplied mile_high_stays.csv and client_properties.csv. Documents and justifies missing-value treatment, unusual values, derived variables, and exclusions, including their effects on counts. Submitted code reproduces the results in a fresh session.
3	Uses the supplied files, explains the main preparation decisions, and reproduces the main results. Minor documentation gaps do not conceal material data issues or prevent execution.
2	Leaves important preparation choices unexplained, mishandles some data, or depends on manual steps or session state that impede reproduction.
1	Has major unresolved data problems, unexplained exclusions, or code that cannot reproduce the principal results from the supplied files.

Technical report communication - 15 points

Rating	Evidence in the submission
4	Provides a complete rendered Word or PDF document with all analysis code, relevant output, and plots. Clear sections, useful comments, and prose let an unfamiliar technical colleague understand what was done, why, and what it means.
3	Provides a readable Word or PDF report containing the principal code, output, and plots. Most steps and results are explained; a few omissions or unclear passages require modest reader effort.
2	Omits important code, output, or explanation, or relies mainly on code comments and unexplained results. Organization makes the workflow difficult to follow.
1	Provides a substantially incomplete or unusable report, omits the rendered report deliverable, or gives too little explanation to understand the work.

What reproduction should establish

Using the supplied files and submitted code in order should recreate the reported analyses, predictions, and plots without undocumented interactive work. Necessary packages and derived-variable rules should be evident.

Executive summary

Executive summary - 15 points

Rating	Evidence in the submission
4	Fits on one page and stands alone for a manager. Descriptive headings organize the objective, approach, key findings and implications, and recommendation. Specific numbers and material limitations support clear plain-language decisions; detailed procedures and code are omitted.
3	Fits on one page and covers all four content areas with understandable findings, supporting numbers, and a recommendation. Headings, concision, or the balance of content need minor improvement.
2	Exceeds one page, omits a required content area or necessary evidence, relies on outside material for key findings, or includes distracting technical detail.
1	Is substantially incomplete, unclear, or too technical to communicate the business problem, main evidence, and proposed action to a manager.